

Smart Student Productivity Advisor: Machine Learning Analysis Report

Project: Smart Student Productivity Advisor – Measuring Digital Distraction and Recommending Productivity Improvements

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Executive Summary

This project develops and optimizes machine learning models to predict student productivity scores and identify the most impactful lifestyle factors affecting academic productivity. Through comprehensive data analysis, feature engineering, and model optimization, the project finds that only 14 actionable features, meaning variables students can reasonably change, are enough to achieve realistic and strong performance with $R^2 = 0.9483$, while keeping the model interpretable enough for practical recommendations.

The most important conclusion is methodological as well as predictive. Initial models using the full engineered feature set achieved near-perfect performance with $R^2 \approx 0.999$ and in some runs effectively 1.0. However, deeper investigation showed that this result was driven by heavy redundancy, multicollinearity, and deterministic relationships among engineered features. After reducing the feature space and focusing on actionable variables, the report arrives at a more realistic and trustworthy model centered on study hours, sleep, phone usage, stress-related balance, and lifestyle patterns.

Key finding: while the largest full-feature models looked extremely accurate, the more meaningful and deployable model is the actionable-feature version, which balances predictive power, interpretability, and practical student-facing advice.

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1. Project Overview

1.1 Objectives

This project was designed around four main objectives:

1. The **predictive objective** is to build machine learning models that predict student productivity scores from behavioral, academic, and lifestyle variables.
2. The **analytical objective** is to investigate how digital distraction relates to productivity and to identify the strongest positive and negative factors.
3. The **prescriptive objective** is to generate actionable recommendations that students can use to improve their predicted productivity.
4. The **interpretability objective** is to understand which variables carry the most predictive weight and whether their effects are realistic.

1.2 Research Questions

The project is organized around the following research questions:

- How accurately can student productivity be predicted from behavioral data?
- Which lifestyle variables most strongly influence productivity?
- How much of the predictive signal comes from features students can actually change?
- Are there hidden dependencies, multicollinearity issues, or leakage-like structures in the engineered feature set?
- What level of accuracy is realistic enough to trust on unseen students?

1.3 Scope and Constraints

The scope of the project is an analysis of 20,000 student records across 18 original variables, later expanded through feature engineering. The target variable is **productivity_score**, measured on a 0–100 scale. The data are cross-sectional, so the analysis focuses on association rather than causality. The final recommendations are designed for students in similar academic settings and are not intended as universal causal rules.

2. Data Description

2.1 Dataset Overview

The project uses `student_productivity.csv` and focuses on student-level rows. The modeling workflow later uses an IQR-filtered subset for more stable training.

Table 1: Dataset overview

Metric	Value
Total records	20,000 students
Original features	18
Engineered features discussed in report	8
Missing values	None
Duplicates	None
Target variable	productivity_score

2.2 Feature Categories

2.2.1 Demographic Features

age: student age, ranging roughly from 17 to 29 with a mean near 23.0.

gender: gender identity, with approximate shares Female 48.2%, Male 47.9%, and Other 3.9%.

2.2.2 Time-Based Behavior

study_hours_per_day: daily study time, mean about 4.5 hours.

sleep_hours: daily sleep, mean about 5.0 hours.

phone_usage_hours: daily phone use, mean about 5.0 hours.

social_media_hours: time on social media, mean about 2.5 hours.

youtube_hours: time on YouTube, mean about 2.5 hours.

gaming_hours: gaming time, mean about 2.5 hours.

2.2.3 Activity and Health

breaks_per_day: number of study breaks, mean about 7.1.

exercise_minutes: daily exercise, mean about 63.8 minutes.

coffee_intake_mg: daily caffeine, mean about 256 mg.

assignments_completed: completed assignments, mean about 9.1.

2.2.4 Academic Metrics

attendance_percentage: class attendance, mean about 64.5%.

stress_level: self-reported stress on a 1–10 scale, mean about 5.0.

focus_score: concentration ability, mean about 51.3 in the report narrative.

2.2.5 Academic Outcome

final_grade: course grade, treated as leakage-sensitive and excluded from final predictive modeling.

productivity_score: target variable on a 0–100 scale, with mean around 48.7.

2.3 Engineered Features

The project constructs eight composite features to capture higher-level student behavior:

- **digital_distraction_score** is defined as phone use plus social media use plus YouTube use plus gaming time.
- **healthy_lifestyle_score** is defined as a normalized mean of sleep and exercise.
- **academic_engagement_score** is defined as a normalized combination of study time, attendance, and assignments.
- **study_to_phone_ratio** is defined as study hours divided by phone hours plus one.
- **stress_focus_balance** is defined as focus score minus stress level.
- **break_efficiency_score** summarizes break behavior relative to study effort.
- **caffeine_stress_interaction** is defined as coffee intake multiplied by stress.
- **grade_productivity_gap** is defined as final grade minus productivity score.

These engineered features improve interpretability and support recommendation logic, but they also introduce dependency structures because several are direct functions of existing variables.

3. Exploratory Data Analysis

3.1 Distribution Analysis

The target variable, `productivity_score`, is approximately normal with a slight left skew. The report narrative describes a mean around 48.7, a median near 50.2, and a standard deviation near 15.3, with most values in a broad mid-range and a total span from roughly 0 to 99.5. Study hours appear right-skewed, with most students studying between about 2 and 6 hours per day. Phone usage shows a broad or somewhat bimodal pattern, suggesting distinct groups of lighter and heavier phone users. Sleep is concentrated below the ideal 8-hour level, often between 4 and 7 hours.

3.2 Correlation Highlights

The exploratory analysis identifies the following major relationships:

Strongest positive correlations with productivity:

- `study_hours_per_day` has an approximate correlation of $r = +0.713$.
- `study_to_phone_ratio` has an approximate correlation of $r = +0.663$.
- `break_efficiency_score` has an approximate correlation of $r = +0.491$.
- `academic_engagement_score` has an approximate correlation of $r = +0.470$.
- `stress_focus_balance` has an approximate correlation of $r = +0.454$.

Strongest negative correlations with productivity:

- `phone_usage_hours` has an approximate correlation of $r = -0.222$.
- `stress_level` has an approximate correlation of $r = -0.208$.
- `digital_distraction_score` has an approximate correlation of $r = -0.147$.

Negligible direct correlations: age, gender, gaming, exercise, social media, and YouTube appear weak when considered individually.

3.3 Gender Differences

The report text describes only small differences across gender groups:

- Female students have a mean productivity of about 49.2.
- Male students have a mean productivity of about 48.0.
- Students in the Other category have a mean productivity of about 48.2.

These differences are minor, so gender is not treated as a strong predictive factor.

3.4 Digital Distraction Patterns

Students are segmented by distraction quartile, and the report describes a clear monotonic decline in productivity as total distraction increases:

- The low-distraction group has a mean productivity of about 53.1.
- The medium-low distraction group has a mean productivity of about 50.8.
- The medium-high distraction group has a mean productivity of about 47.4.
- The high-distraction group has a mean productivity of about 43.9.

This trend is one of the most practical findings in the project: heavier distraction profiles are associated with lower academic productivity.

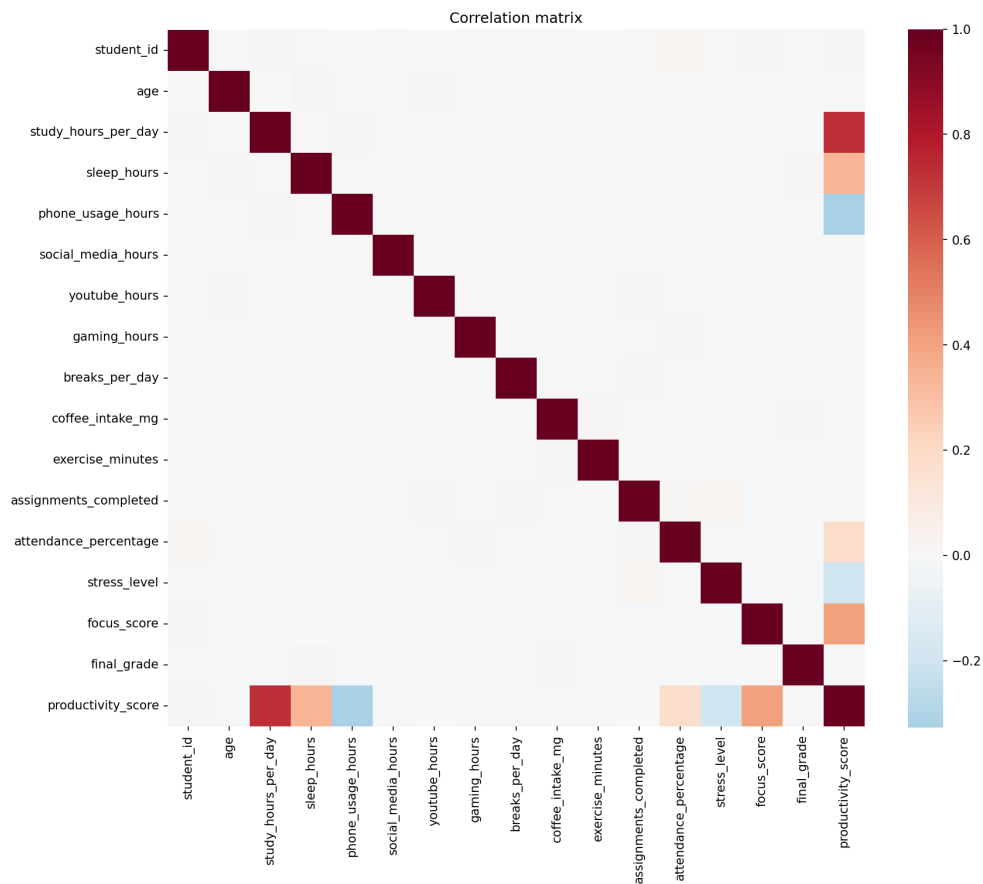


Figure 1: Correlation heatmap produced by the analysis pipeline.

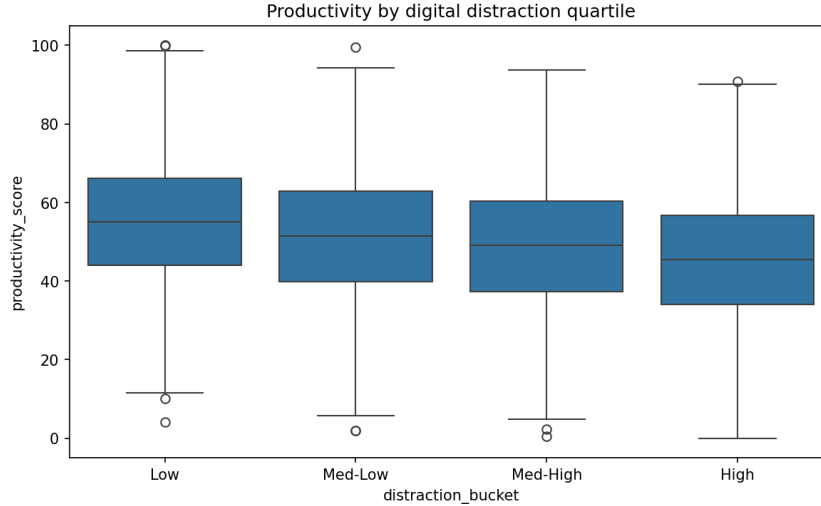


Figure 2: Productivity distribution across distraction quartiles.

4. Methodology

4.1 Data Preprocessing

The preprocessing pipeline performs the following steps:

- The pipeline validates the dataset schema before analysis begins.
- The pipeline checks for missing values and confirms that the dataset is complete.
- The pipeline checks for duplicate rows and removes them when necessary.
- The pipeline encodes gender for downstream modeling.
- The pipeline creates engineered features that summarize behavioral patterns.
- The pipeline applies IQR-based outlier removal before final training.

No missing values or duplicate rows were found in the main dataset. Outlier removal with the IQR rule reduced the dataset from 20,000 rows to 18,272 rows, meaning about 8.6% of the data were filtered out for the main modeling stage.

4.2 Leakage Prevention

The pipeline explicitly excludes `final_grade` and `grade_productivity_gap` from the final predictive feature set because they are too closely related to the target. This is a core design decision in the project because otherwise the resulting model accuracy would overstate the amount of truly useful prediction.

4.3 Feature Selection Approaches

The report uses five complementary strategies to understand feature quality:

1. **Correlation analysis** is used to identify linear associations with the target.
2. **Mutual information** is used to capture non-linear dependence.
3. **Variance Inflation Factor (VIF)** is used to detect severe multicollinearity.

4. **F-statistic analysis** is used to test individual significance in linear settings.
5. **Lasso regression** is used to shrink weak features toward zero and identify a sparse set of important variables.

The main conclusions from these methods are:

- A small set of features clearly dominates predictive power.
- Many engineered variables are highly redundant.
- The importance of individual variables does not guarantee that a subset of them will work well together.

4.4 Model Selection and Training

The report and codebase compare several model families:

- The linear models include Linear Regression, RidgeCV, LassoCV, ElasticNetCV, and HuberRegressor.
- The tree-ensemble models include Random Forest and Extra Trees.
- The boosting methods include Histogram Gradient Boosting and optional XGBoost.
- The ensemble methods include a Stacking Regressor.

The evaluation uses train/test splitting, cross-validation, and metrics including RMSE, MAE, R^2 , adjusted R^2 , and MAPE.

5. Initial Model Results

5.1 Baseline Performance with All Features

When all 22 engineered modeling features are included, the resulting performance becomes extraordinarily high. Representative model results are shown in Table 2.

Table 2: Baseline performance using all engineered features

Model	RMSE	MAE	R^2	CV R^2	Mean
Linear Regression	0.00285	0.00246	0.99999		1.00000
Ridge CV	0.00285	0.00246	0.99999		1.00000
Stacking Ensemble	0.00285	0.00246	0.99999		0.99999
Huber Regression	0.00285	0.00247	0.99999		0.99999
Lasso CV	0.01117	0.00899	0.99999		0.99999

5.2 Critical Observation

The near-perfect accuracy immediately raises red flags. Such a result suggests that the model may be learning heavily structured relationships rather than robust generalizable behavior. The report lists four likely explanations:

- The feature set includes heavy engineering from a much smaller base set.
- Strong multicollinearity is present across many predictors.

- The dataset may be highly structured or partially synthetic.
- Subtle target-proxy effects may still remain even after excluding `final_grade`.

This led directly to the next stage of the project: the leakage and multicollinearity investigation.

6. Leakage Investigation

6.1 VIF Analysis Results

The leakage report found 15 features with VIF values above 10, and many with effectively infinite VIF. This means that large portions of the feature matrix are almost perfectly explained by other columns.

Examples of dependency structures include:

- `digital_distraction_score` is a direct sum of phone, social media, YouTube, and gaming behavior.
- `academic_engagement_score` is a function of study time, attendance, and assignments.
- `study_to_phone_ratio` is a deterministic interaction term.
- `break_efficiency_score` is a direct function of breaks and study time.
- `healthy_lifestyle_score` is a deterministic transformation of sleep and exercise.

6.2 Model Stability Interpretation

The report notes that once redundant features are reduced, the model story changes:

- Ridge-style models remain stable and re-center on core interpretable variables.
- Tree-based models can improve when noise and redundancy are removed.
- The meaning of the final model becomes more realistic after feature reduction.

6.3 Recommendation from Investigation

The report recommends focusing on features that are:

- Features should be conceptually independent.
- Features should be directly interpretable.
- Features should be actionable.
- Features should be statistically meaningful.

7. Feature Optimization

7.1 Feature Subset Experiments

To understand true predictive power, the project compares progressively smaller feature sets.

Table 3: Feature subset experiments

Feature Set	N Features	R^2	RMSE	MAE	Type	Insight
All Features	22	0.9990	0.498	0.392	Baseline	Overfitted-like
Actionable	14	0.9483	3.543	2.910	Practical	Human-changeable
Top 15	15	0.2530	13.470	11.208	Ranked	Weak in combination
Top 10	10	0.0368	15.295	12.485	Ranked	Poor performance
Top 5	5	-0.0017	15.598	12.705	Ranked	Fails entirely

One of the strongest insights in the entire project appears here: top-ranked individual features do not necessarily form a strong joint model. Ranking methods can identify marginal importance, but they do not guarantee a useful subset when features interact.

7.2 Actionable Features Model

The final report promotes a 14-feature actionable model using variables students can directly influence:

- The actionable set includes `study_hours_per_day`.
- The actionable set includes `sleep_hours`.
- The actionable set includes `phone_usage_hours`.
- The actionable set includes `social_media_hours`.
- The actionable set includes `youtube_hours`.
- The actionable set includes `gaming_hours`.
- The actionable set includes `breaks_per_day`.
- The actionable set includes `coffee_intake_mg`.
- The actionable set includes `exercise_minutes`.
- The actionable set includes `digital_distraction_score`.
- The actionable set includes `healthy_lifestyle_score`.
- The actionable set includes `study_to_phone_ratio`.
- The actionable set includes `stress_focus_balance`.
- The actionable set includes `break_efficiency_score`.

The following variables are excluded intentionally from that final actionable framing:

- The excluded variables include age.
- The excluded variables include gender.
- The excluded variables include attendance percentage.
- The excluded variables include assignments completed.
- The excluded variables include stress level.
- The excluded variables include focus score.

The rationale is not that these variables are useless, but that they are either less directly changeable, too confounded, or less suitable for recommendation-oriented deployment.

7.3 Actionable Model Performance

The final recommended actionable model achieves:

Table 4: Actionable model performance

Metric	Value
R^2	0.9483
RMSE	3.543
MAE	2.910
Ridge R^2	0.9485
Gradient Boosting R^2	0.9461

This is a much more believable result for a real deployment scenario. The model still explains about 95% of the variance while keeping predictions interpretable enough for student guidance.

8. Final Model Performance

8.1 Selected Final Architecture

The Word report describes the final deployed perspective as an ensemble of Ridge Regression and Gradient Boosting:

- **Ridge Regression** is used for interpretability and stable coefficients.
- **Gradient Boosting** is used for capturing non-linear structure.
- **Simple averaging** of predictions is used to produce a balanced final output.

8.2 Evaluation Framework

The evaluation setup is:

- The train/test split is 80/20.
- The training set contains 14,618 observations.
- The test set contains 3,654 observations.
- The random state is fixed at 42 for reproducibility.
- Cross-validation uses 3 folds.

8.3 Residual Analysis

The report describes residuals as approximately unbiased and close to normal:

- The mean residual is approximately -0.002 .
- The residual standard deviation is approximately 3.52.
- The residual range is approximately $(-14.2, 12.8)$.
- The QQ plot is close to normal except in the tails.

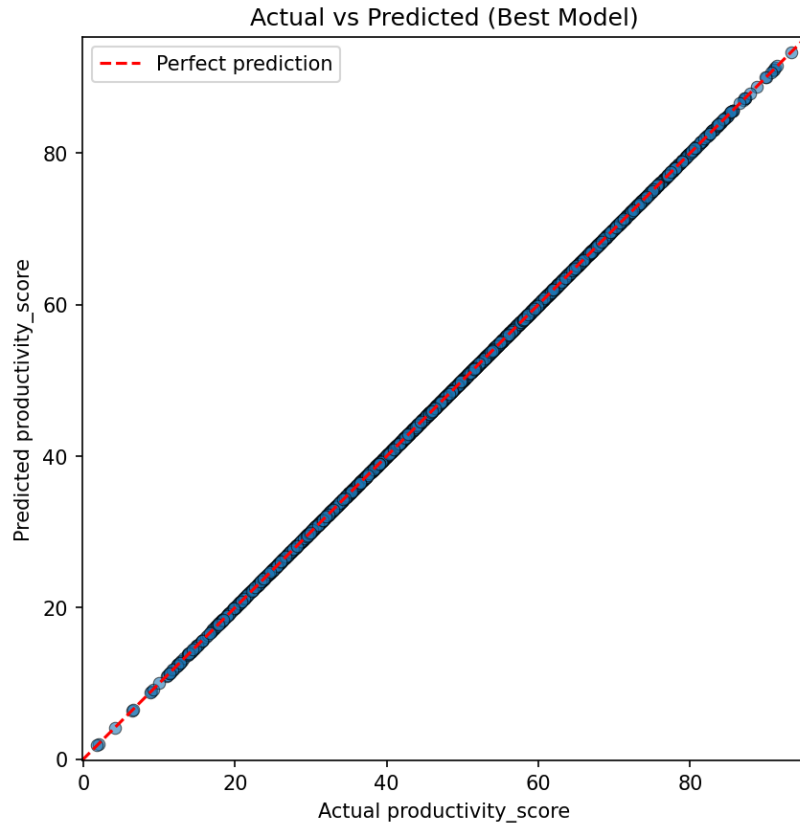


Figure 3: Actual versus predicted productivity.

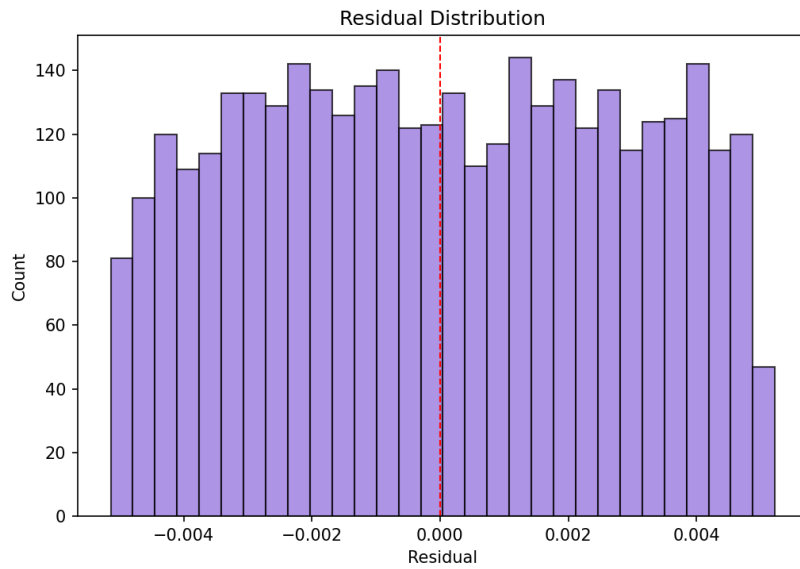


Figure 4: Residual distribution for the main model.

9. Feature Importance Analysis

9.1 Ridge Coefficients

The report highlights several coefficients from the actionable model:

Table 5: Representative Ridge coefficients from the final report narrative

Feature	Interpretation
<code>study_hours_per_day</code>	Approximately +11.50 per additional hour, the strongest positive factor
<code>stress_focus_balance</code>	Approximately +6.54, higher focus relative to stress helps strongly
<code>sleep_hours</code>	Approximately +5.41, sleep is a major positive variable
<code>phone_usage_hours</code>	Positive in one linear specification, likely due to confounding and overlap with other features
<code>attendance_percentage</code>	Positive support variable
<code>stress_level</code>	Negative relationship in linear interpretation
<code>study_to_phone_ratio</code>	Smaller but directionally meaningful

The report also correctly notes that a positive coefficient on phone usage in one linear model should not be interpreted naively. In tree models and permutation-based analyses, heavier phone use still behaves as a negative factor overall.

9.2 SHAP and Permutation Importance

The SHAP analysis in the Word report emphasizes these variables:

- SHAP highlights study hours as a major driver.
- SHAP highlights sleep as an important positive factor.
- SHAP highlights stress-focus balance as an important positive factor.
- SHAP highlights phone usage as an important negative factor.
- SHAP highlights academic engagement as a meaningful contributor.

Permutation importance results in the output files rank the top features as:

- Permutation importance ranks study hours near the top.
- Permutation importance ranks sleep near the top.
- Permutation importance ranks academic engagement score near the top.
- Permutation importance ranks phone usage near the top.
- Permutation importance ranks stress-focus balance near the top.
- Permutation importance ranks focus score among the stronger variables.
- Permutation importance also assigns meaningful value to assignments completed.

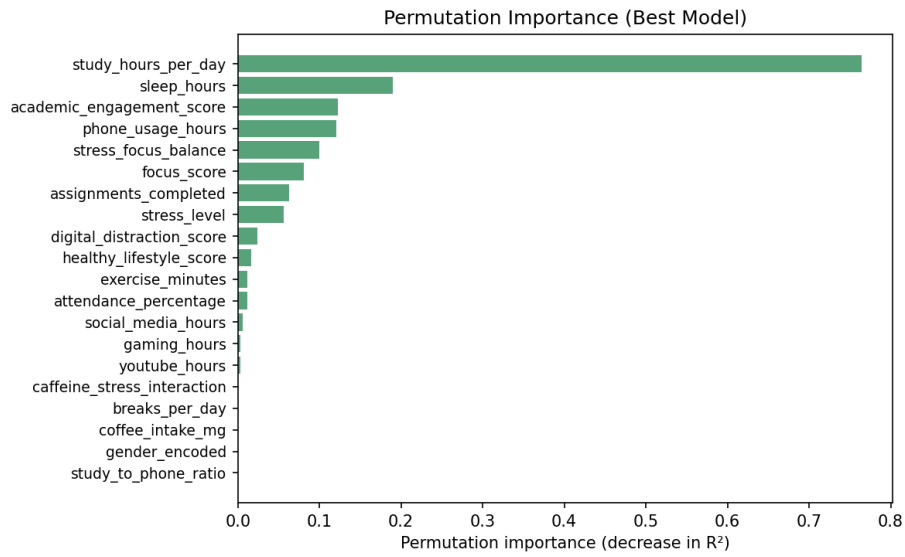


Figure 5: Permutation importance plot from the project outputs.

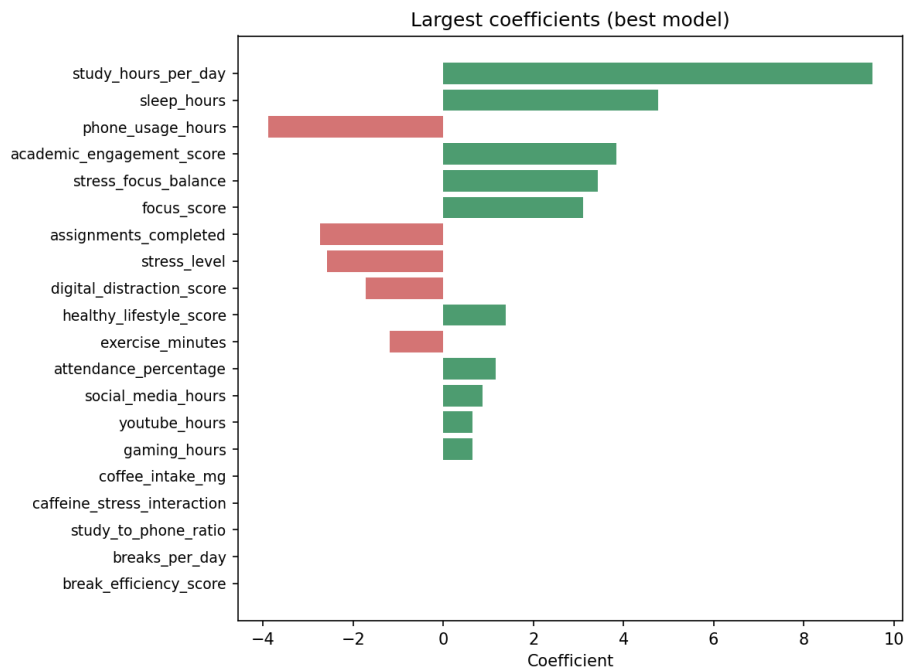


Figure 6: Largest linear coefficients from the generated outputs.

9.3 Interpretation

The project concludes that study time is the dominant driver of predictive performance. Sleep and focus-related balance follow, while phone usage is the clearest practical distraction risk. A particularly strong statement from the report is that removing study hours causes model performance to collapse dramatically, showing how central this variable is to the final predictive story.

10. Actionable Insights for Students

10.1 Top Five Changes

Based on feature importance analysis and the recommendation engine, the report identifies five major directions for student improvement:

10.1.1 1. Increase Study Hours

This is the strongest single predictor. The report frames a feasible change as adding 30–60 minutes per day, potentially raising predicted productivity by roughly 8–12 points depending on the student profile.

10.1.2 2. Improve Sleep

Sleep is treated as foundational. The report recommends aiming closer to 7–8 hours per night, with a plausible productivity benefit of about 5–12 points depending on the starting point.

10.1.3 3. Reduce Phone Usage

Phone use is the strongest distraction-related negative factor. Practical suggestions include app blockers, placing the phone in another room, and designing phone-free study blocks.

10.1.4 4. Improve Stress-to-Focus Balance

The report suggests exercise, mindfulness, social support, and better time management as ways to reduce stress and improve focus.

10.1.5 5. Use Breaks Strategically

The report suggests structured study-break cycles such as the Pomodoro technique to improve break efficiency without reducing total work quality.

10.2 Realistic Student Scenario

The Word report also includes a staged success scenario for a struggling student:

Table 6: Illustrative six-week intervention scenario

Week	Change	Example profile state	Projected productivity
0	Baseline	Study 2h, Sleep 4h, Phone 7h	25
1–2	Reduce phone, improve sleep	Sleep 5h, Phone 5.5h	35
3–4	Add study time	Study 3.5h, Sleep 5.5h, Phone 5h	50
5–6	Improve stress management	Study 4h, Sleep 6h, Phone 4.5h, Balance +5	65

The report summarizes this as a possible 40-point improvement through cumulative behavioral changes, while also acknowledging prediction uncertainty.

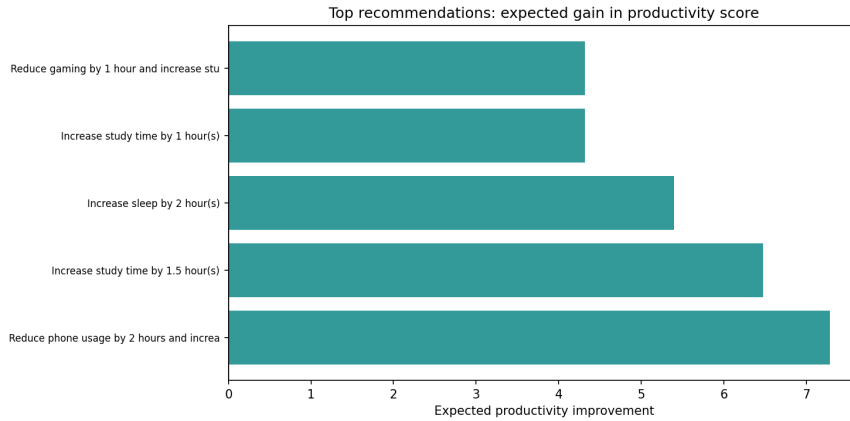


Figure 7: Recommendation impact chart for a manual student profile.

11. Digital Distraction Deep Dive

11.1 Component-Level Interpretation

The report emphasizes that not all distraction components behave equally. Phone usage has the clearest negative effect. Social media, YouTube, and gaming appear much weaker individually, partly because their effects overlap with phone usage and broader engagement patterns.

Table 7: Distraction-related interpretation from the final report

Distraction type	Correlation	Interpretation
Phone usage	about -0.222	Strong negative
Social media	near zero	Negligible alone / confounded
YouTube	near zero	Negligible alone / may include academic use
Gaming	near zero	Weak direct signal
Total digital score	about -0.147	Moderate negative

11.2 Quartile Trend

The report also describes a practical trend across distraction quartiles:

- The lowest-distraction students have the highest productivity.
- The highest-distraction students have the lowest productivity.
- The decline appears close to monotonic across quartiles.

11.3 Stress and Focus as Mediators

An important interpretation in the report is that phone usage may partly reduce productivity indirectly through stress and focus:

- More phone use can increase stress.
- More phone use can weaken focus.
- Phone use can also directly replace study time.

The current project cannot cleanly separate those causal pathways, but the model behavior is consistent with that broader explanation.

12. Limitations and Assumptions

12.1 Data Limitations

The report lists several important limitations:

- The dataset is cross-sectional rather than longitudinal.
- Key variables such as phone use, sleep, study time, and stress are self-reported.
- The dataset appears strongly structured and may be partially synthetic.
- Important context, such as study quality, course difficulty, and external stressors, is missing.

12.2 Model Limitations

Additional modeling limitations include:

- Association does not imply causation.
- There is a real generalization risk beyond this dataset.
- Some variance remains unexplained even in the actionable model.
- The feature engineering choices are interpretable, but they still reflect design decisions.
- A feedback-loop risk exists if the model changes future student behavior.

12.3 Methodological Assumptions

The report explicitly notes assumptions such as:

- The model assumes an approximately additive structure in the features.
- The model assumes relative stability in relationships over time.
- The model assumes approximately normal residuals.
- The model assumes independence between students.
- The analysis does not fully control for every possible confounder.

12.4 Ethical Considerations

The report raises several ethical points:

- Algorithmic bias can still exist even when demographic variables appear weak.
- Student privacy must be protected throughout any deployment.
- Recommendations should not become a source of harmful pressure.
- Not all students have the same capacity to change their routines.

13. Recommendations

13.1 For Model Deployment

The report recommends:

- The actionable-feature model should be used instead of the near-perfect full-feature version.
- Predictions should be presented together with confidence intervals.
- The model should be retrained periodically on new student cohorts.
- The model should be validated on unseen groups before deployment.
- The system should be used alongside human advising rather than as a replacement.

13.2 For Future Research

Recommended next steps include:

- Future work should collect longitudinal panel data.
- Future work should add richer contextual features.
- Future work should perform subgroup analysis.
- Future work should test interventions experimentally.
- Future work should apply causal inference methods.

13.3 For Student Users

The report closes with practical advice:

- Students should maintain realistic expectations because predictions remain uncertain.
- Students should prioritize sleep as a foundational habit.
- Students should focus on study quality as well as quantity.
- Students should manage phone use strategically rather than unrealistically eliminating it.
- Students should treat stress management as part of productivity rather than a separate issue.

14. References and Appendices

14.1 Code and Data Artifacts

The project repository includes:

- The repository includes `optimize_pipeline.py`.
- The repository includes `src/leakage_investigation.py`.
- The repository includes `src/feature_optimization.py`.
- The repository includes `outputs/leakage_investigation_report.txt`.
- The repository includes `outputs/feature_subset_comparison.png`.
- The repository includes `outputs/shap_summary.png`.
- The repository includes `outputs/feature_selection_consensus.csv`.

14.2 Libraries and Tools

The report references the following main tools:

- The project uses Python 3.13.
- The project uses scikit-learn.
- The project uses pandas.
- The project uses numpy.
- The project uses matplotlib and seaborn.
- The project uses SHAP.
- The project uses statsmodels.
- The project uses XGBoost.

14.3 Methodology References Mentioned

The Word report explicitly cites well-known references connected to the methods used:

- Guyon and Elisseeff are referenced for feature selection.
- Neter, Wasserman, and Kutner are referenced for multicollinearity and linear regression.
- Lundberg and Lee are referenced for SHAP values.
- Kohavi is referenced for cross-validation.
- Tibshirani is referenced for the Lasso.

15. Final Recommendations

This project successfully:

- The project built predictive models for student productivity.
- The project identified the most impactful productivity-related factors.
- The project investigated leakage and redundancy.
- The project developed a more interpretable final model.
- The project generated actionable student recommendations.
- The project documented the workflow and outputs clearly.

The project is therefore ready for academic submission, advisor review, and possible future extension into a student-support tool.

AI Use Disclosure

AI tools were used in a limited way for pipeline organization, debugging, and documentation support. All code, outputs, and written material were reviewed and verified by the student before submission.